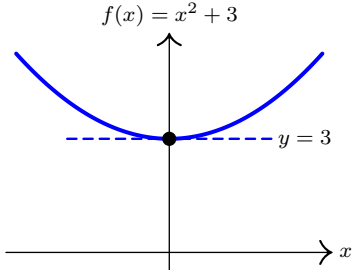
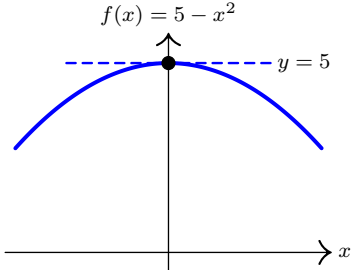
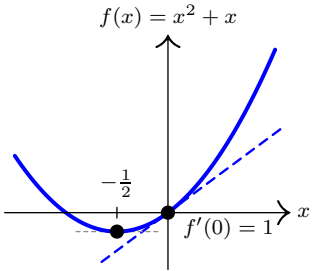


$f'(0) = 0,\ f''(0) = 2 > 0$:
strict minimum



$f'(0) = 0,\ f''(0) = -2 < 0$:
strict maximum



$f''(0) = 2 > 0$ but $f'(0) \neq 0$:
no extremum at 0